



DataCo Global

Phase 2 | Source System Understanding & Data Lineage



Source Systems Identified	2 (inferred: ERP / Order Management, E-commerce Clickstream Tracker)
Datasets Cataloged	2 (DataCoSupplyChainDataset.csv, tokenized_access_logs.csv)
Data Boundaries	Transactional order data (2015–2018, inferred) and web session data

1. Purpose

This document identifies and describes the source systems that produce the data used in this analytics project. It defines the boundaries of each system, the data they contribute, and the relationships between datasets.

Understanding the source architecture is a prerequisite for meaningful data analysis. Without this context, analysts risk misinterpreting field meanings, joining tables incorrectly, or drawing conclusions unsupported by the data.

2. Source System Overview

Two distinct source systems contribute data to this project. Based on the structure and content of the datasets, the following system types are inferred.

System	Inferred Type	Data Provided	Technology Indicators
System A	ERP / Order Management System (inferred)	All transaction data: orders, customers, products, shipping, financials	Structured relational schema; denormalized export for analytics; consistent key naming conventions (inferred)
System B	E-commerce Clickstream Tracker (inferred)	Web session data: page views, navigation paths, product views	Event-log format; different naming convention (snake_case); higher row count suggests automated logging (inferred)

Note: System types are inferred from dataset characteristics. No direct access to source systems or documentation was available.

3. System A: ERP / Order Management

This system is the primary data source, contributing all transactional data including orders, customers, products, shipping, and financial metrics. The data is provided as a denormalized flat-file export, which is typical for analytics extracts from operational ERP systems.

Dataset Summary

Property	Value
Source File	DataCoSupplyChainDataset.csv
Rows	~180,519
Columns	53
Primary Key	Order Id + Order Item Id (composite)
Foreign Keys	Customer Id, Product Card Id, Category Id, Department Id
Date Range (Inferred)	2015–2018 based on date column values
Known Business Processes	Order-to-Cash (order capture, fulfillment, delivery, payment)

Data Domains

Domain	Field Count	Description
Customer	11	Customer identity and contact information.
Product	11	Product catalog, category, and department data.
Order Transaction	14	Order header and line-item transaction data.
Shipping & Logistics	8	Delivery tracking and shipping destination data.
Financial Metrics	5	Revenue and profitability measures.
Geospatial	2	Latitude and longitude coordinates.

Likely Business Function

The system that generates this data is typically operated by an **Order Management** or **Supply Chain Operations** function. This is inferred from the nature of the data (order processing, inventory, shipping) and standard retail industry practices.

4. System B: E-commerce Clickstream Tracker

This system captures user browsing behavior on the DataCo e-commerce website. Each row represents a user interaction event, with navigation paths encoded in the click_stream field.

Dataset Summary

Property	Value
Source File	tokenized_access_logs.csv
Rows	~415,000
Columns	8
Primary Key	user_id + event_date (composite)
Foreign Keys	item_id → Product Card Id, order_id → Order Id, user_id → Customer Id
Data Type	Event-log format with session tracking
Known Business Processes	Customer browsing, search, product discovery, purchase conversion

Data Fields

Field	Description
event_date	Timestamp of the browsing event.
click_stream	Encoded navigation path during the session.
item_id	Product identifier viewed. Joins to Product Card Id.
user_id	Web user identifier. May join to Customer Id if authenticated.
order_id	Order identifier if the session led to a purchase.
product_name	Name of the product viewed.
category_name	Category of the product viewed.
access_token	Tokenized session identifier.

Likely Business Function

Clickstream tracking is typically managed by **Digital / E-commerce** or **Web Analytics** teams. This is inferred from the nature of the data and standard industry practice.

5. Cross-System Relationships

The two datasets can be joined through the following field mappings. These relationships enable cross-domain analysis.

Clickstream Field	Transaction Field	Relationship	Notes
item_id	Product Card Id	Many-to-One	Product views per product. Enables browse-to-purchase analysis.
order_id	Order Id	One-to-One	Links session to order when conversion occurs.
user_id	Customer Id	Many-to-One	Only valid for authenticated sessions. Anonymous sessions have null or placeholder user IDs.

The **item_id** to **Product Card Id** join is the most reliable cross-dataset relationship. The **order_id** relationship only exists for sessions that ended in a purchase. The **user_id** to **Customer Id** relationship requires authentication status verification.

6. Data Boundaries & Limitations

The following limitations affect how this data can be interpreted:

Limitation	Impact	Mitigation
Denormalized flat-file export	Repeated values across rows (e.g. Sales per customer repeated per line item). Requires careful aggregation.	Confirm aggregation logic before reporting totals.
No access to source system	Cannot verify referential integrity or data transformation logic.	Treat field relationships as inferred. Validate through data profiling.
Clickstream session context unknown	Encoding of click_stream field is undocumented. Session boundaries unclear.	Analyze click_stream patterns to infer structure.
Date range is inferred	Exact time window of data is not documented.	Use min/max of date column for approximation.

These limitations are typical for analytics projects working with exported data. The Data Profiling phase will quantify the impact of each limitation where possible.

7. Data Flow Summary

The overall data flow follows this pattern:

```
ERP / Order Mgmt (inferred)
|
v
DataCoSupplyChainDataset.csv

E-commerce Clickstream (inferred)
|
v
tokenized_access_logs.csv

+-----+-----+
|
v
Analytics Workspace
(current project scope)
```

End of Operational Architecture Discovery. Proceed to Phase 3: Data Dictionary.